

Weekly Progress Report: Week 13

Mujawwad Project Team

Nov 24th – Nov 30th

Week 13: Research on Model Architectures

Task	Description
Phoneme Recognition R&D	Text-to-Phoneme: Integrated Quran Phonemizer (Tajweed-aware) to generate ideal phoneme sequences from reference Ayahs. Audio-to-Phoneme: Evaluated multiple models including wav2vec2-large-xlsr-53-arabic, Allosaurus (IPA), and wav2vec2-lv-60-espeak. Identified limitations in current pre-trained models regarding Quranic nuances. Algorithm: Successfully tested Needleman-Wunsch algorithm for alignment and comparison of phoneme sequences.
ASR Feasibility Analysis	Conducted deep-dive analysis into ASR architecture (Acoustic Models vs. Language Models). Concluded that standard ASR is unsuitable due to its error-correcting nature (Language Model bias); determined that raw Acoustic Model output is required to accurately capture user pronunciation errors.
Signal Processing Approach (MFCC & DTW)	Formulated a secondary validation strategy comparing User Audio against “Golden Audio.” Defined workflow: 1. Forced Alignment for phoneme timestamping. 2. Segmentation of audio per phoneme. 3. MFCC Extraction for spectral representation. 4. DTW (Dynamic Time Warping) calculation to threshold correct/incorrect pronunciation.